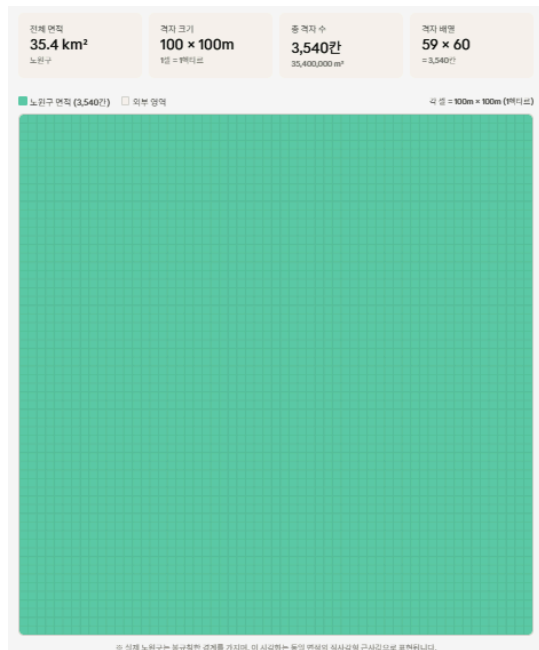
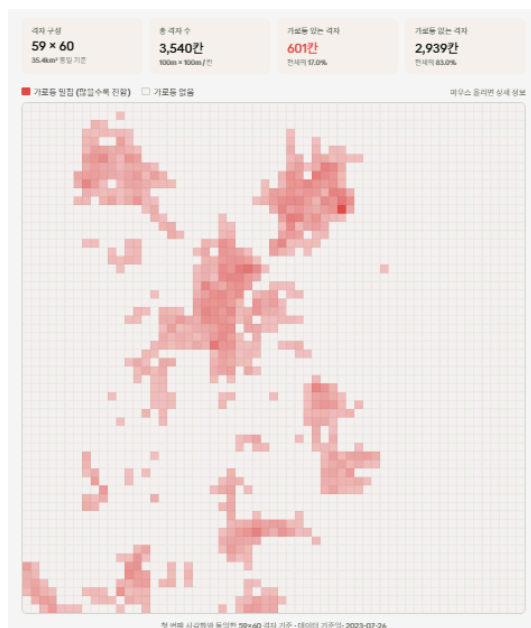


손주영

노원구 100m×100m 기본 격자 생성



기본 격자에 보안등 데이터 결합



파이썬 이용한 시각화 과정 - 코드

```
관리자: C:\Windows\System32\cmd.exe
Microsoft Windows [Version 10.0.26200.8246]
(c) Microsoft Corporation. All rights reserved.

C:\Users\82102\OneDrive\바탕 화면\새 폴더 (2)>python nowon_streetlight_grid.py
격자에 포함된 보안등: 4090개
범위 밖 또는 오류 데이터: 1780개
가로등 있는 격자: 601칸 / 전체 3540칸

가로등 없는 격자: 601칸 (17.0%)
가로등 있는 격자: 2939칸 (83.0%)

보안등 밀집 상위 5개 격자:
12월37 → 40개
19월26 → 26개
11월37 → 25개
28월22 → 24개
08월34 → 23개

C:\Users\82102\OneDrive\바탕 화면\새 폴더 (2)>
```

```
관리자: C:\Windows\System32\cmd.exe
Downloading pyparsing-3.3.2-py3-none-any.whl.metadata (5.8 kB)
Requirement already satisfied: python-dateutil<=2.7 in C:\Users\82102\AppData\Local\Programs\Python\Python314\Lib\site-packages (from matplotlib) (2.9.0.post0)
Requirement already satisfied: six>=1.5 in C:\Users\82102\AppData\Local\Programs\Python\Python314\Lib\site-packages (from python-dateutil<=2.7->matplotlib) (1.17.0)
Downloading matplotlib-3.10.9-cp314-cp314-win_amd64.whl (8.3 MB)
8.3/8.3 MB 1.1 MB/s 0:00:07
Downloading contourpy-1.3.3-cp314-cp314-win_amd64.whl (232 kB)
Downloading cycler-0.12.1-py3-none-any.whl (8.3 kB)
Downloading fonttools-4.63.0-cp314-cp314-win_amd64.whl (2.3 MB)
2.3/2.3 MB 1.2 MB/s 0:00:02
Downloading kiwisolver-1.5.0-cp314-cp314-win_amd64.whl (75 kB)
Downloading packaging-26.2-py3-none-any.whl (100 kB)
Downloading pillow-12.2.0-cp314-cp314-win_amd64.whl (7.2 MB)
7.2/7.2 MB 1.1 MB/s 0:00:06
Downloading pyparsing-3.3.2-py3-none-any.whl (122 kB)
Installing collected packages: pyparsing, pillow, packaging, kiwisolver, fonttools, cycler, contourpy, matplotlib
Successfully installed contourpy-1.3.3 cycler-0.12.1 fonttools-4.63.0 kiwisolver-1.5.0 matplotlib-3.10.9 packaging-26.2 pillow-12.2.0 pyparsing-3.3.2

[notice] A new release of pip is available: 26.0.1 -> 26.1.1
[notice] To update, run: python.exe -m pip install --upgrade pip

C:\Users\82102\OneDrive\바탕 화면\새 폴더 (2)>
```

파이썬 이용한 시각화 결과

